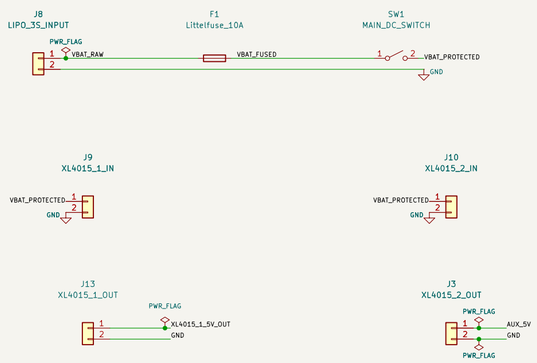
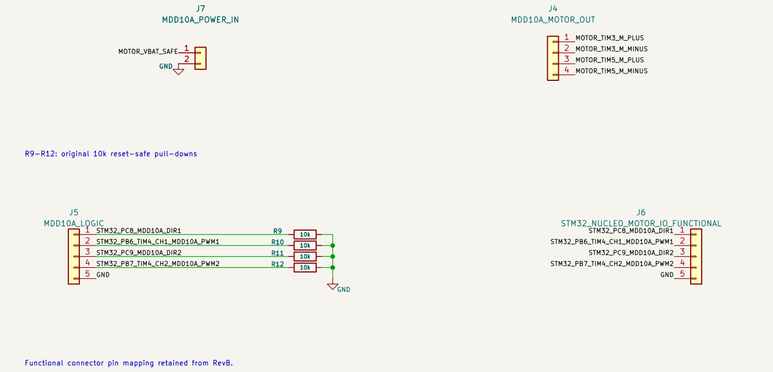
TRACKED MOBILE ROBOT | MVP FUNCTIONAL WIRING | RevB – 2026-11-07

Layout: 2026-11-06 | Component identities reconciled: 2026-11-07 | Original functional connections retained

01 BATTERY & DC/DC INTERFACES



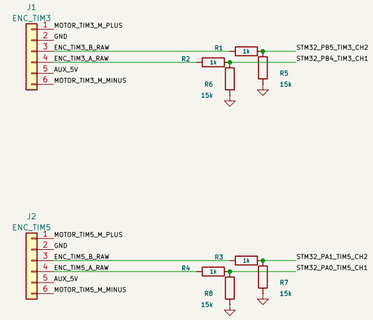
02 MDD10A & STM32 MOTOR I/O



R9–R12: original 10k reset-safe pull-downs

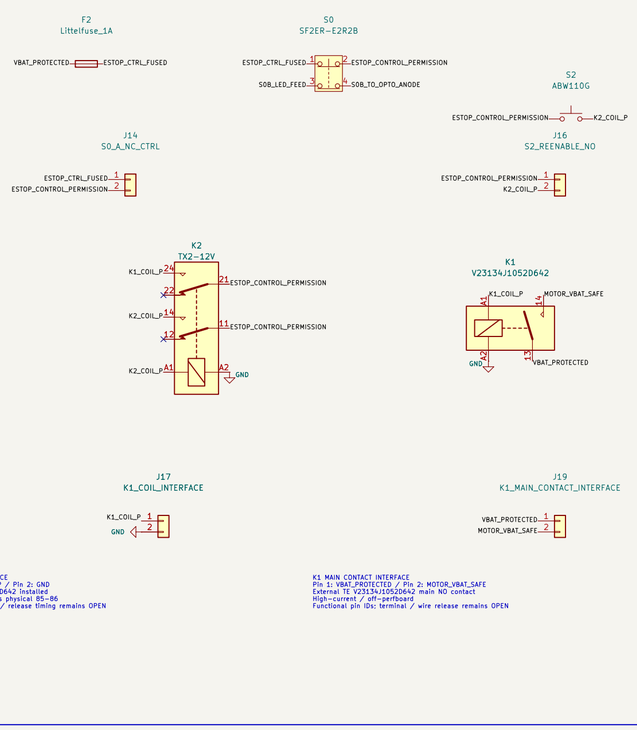
Functional connector pin mapping retained from RevB.

03 DUAL ENCODER CONDITIONING



TIM3 and TIM5 channel identities retained; conditioning values unchanged.

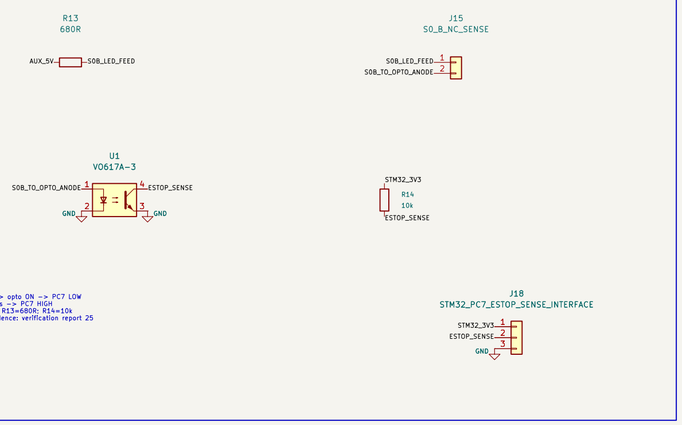
04 E-STOP CONTROL / RELAY INTERFACES



K1 COIL INTERFACE
Pin 1: K1_COIL_P / Pin 2: GND
TE V23134J1052D642 installed
P6KE16CA across physical 85–86
Clamp transient / release timing remains OPEN

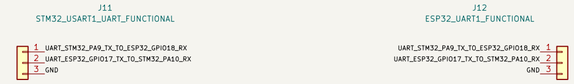
K1 MAIN CONTACT INTERFACE
Pin 1: VBAT_PROTECTED / Pin 2: MOTOR_VBAT_SAFE
External TE V23134J1052D642 main NO contact
High-current / off-perfboard
Functional pin IDs: terminal / wire release remains OPEN

05 E-STOP SENSE / STM32 PC7



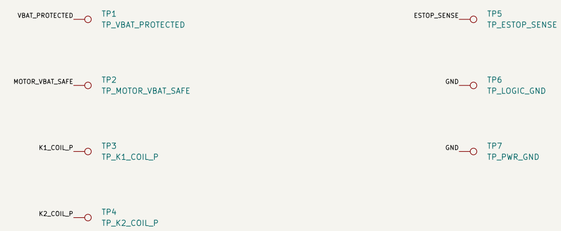
S0-B SENSE
5V/contact closed -> opto ON -> PC7 LOW
pressed/open/5V loss -> PC7 HIGH
Installed: VO617A-3: R13=680R: R14=10k
Conditioned path evidence; verification report 25

06 STM32 – ESP32 UART



115200 8N1 is the original documented link setting.
Matching net names connect TX to the opposite RX; GND is shared.

07 TEST POINTS



08 CURRENT DOCUMENT STATUS

Recorded hardware evidence reconciled; functional diagram, not an as-built harness release.

DOCUMENT STATE – 2026-11-07

UPDATED FROM RECORDED HARDWARE
- S0: Autonics SF2ER-E2R2B; S2: IDEC ABW110G
- K1: TE V23134J1052D642; K2: Panasonic TX2-12V
- F1: Littelfuse 10A; F2: Littelfuse 1A (reported)
- VO617A-3 / 680R / 10k conditioned sense path
- K2 generic-to-physical pin mapping shown below

EVIDENCE BOUNDARIES
- Reports 18 / 19: incoming identification and screens
- Reports 24 / 25 / 30: scoped assembly / bench evidence
- No new physical measurement or firmware change
- F1 series / F2 exact SKU / rating coordination OPEN
- Original functional connector / encoder topology retained
- Overall MVP: PARTIAL / OPEN; T007 BLOCKED

Component labels follow recorded hardware evidence. Functional connectivity is retained.
Use the reconciliation report and dated bench records for physical pin / harness mapping.

09 RELAY MODEL & DRAWING NOTES

K2 CONTROL RELAY | Panasonic TX2-12V installed
SCHEMATIC IDs -> PHYSICAL pins (bottom view)
A1 -> 1 (+); A2 -> 12 (–)
11 -> 9 (COM); 14 -> 8 (NO); HOLD
21 -> 4 (COM); 24 -> 5 (NO); K1 ENABLE
Schematic 12/22: unused NC contacts
Generic symbol is not a physical footprint pin map

48 physical components / original references and connections retained.
Recorded component identities updated; rating acceptance remains separate.
F1: 10A reported. 257 marking / 287 order discrepancy OPEN; F2 exact SKU OPEN.

DOCUMENT RECONCILIATION 2026-11-07
S0 body: SF2ER-E2R2B: ordered suffix -A not visually confirmed
S2: IDEC ABW110G: no new hardware test
Evidence: assets/logs/pcb_design/2026-11-07_cad_closeout

Existing hardware evidence only; rating / as-built release remains open
Component identities reconciled; functional pin IDs retained

Sheet: /
File: Tracked_Mobile_Robot_Wiring_RevB.kicad_sch

Title: Tracked Mobile Robot Wiring

Size: A2	Date: 2026-11-07	**Rev: RevB-DOC1**
KiCad E.D.A. 10.0.5		Id: 1/1